

Lattice QCD Data Summary Report

Chiral Condensates and Susceptibilities Across Finite Temperature

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September 2026

Abstract

This report provides numerical measurement data tables and temperature-dependent figures for: (1) bare and renormalized subtracted chiral condensates $\Delta_{l,s}$, (2) temporal scaling series at $\beta = 4.17$, and (3) light-quark chiral susceptibilities χ_{scaled} [MeV²] computed with unbiased quadratic estimators across finite temperatures.

1 Definitions and Observables

1.1 Chiral Condensates

The single-flavor bare chiral condensate on a lattice volume $V = N_s^3 \times N_t$ is defined as

$$\langle \bar{\psi}\psi \rangle_q = \frac{1}{V} \langle \text{Tr} (D_q^{-1}) \rangle, \quad q \in \{l, s\} \quad (1)$$

evaluated via stochastic Z_2 volume source vectors.

The renormalized subtracted chiral condensate $\Delta_{l,s}$ eliminates additive ultraviolet divergences and residual chiral symmetry breaking:

$$\Delta_{l,s}(T) = \frac{1}{Z_m(\beta)} \left[\langle \bar{\psi}\psi \rangle_l - \frac{m_l + m_{\text{res}}}{m_s + m_{\text{res}}} \langle \bar{\psi}\psi \rangle_s \right] \quad (2)$$

where $m_{\text{res}}(\beta)$ is the residual quark mass, and $Z_m(\beta)$ is the scalar mass renormalization factor. The temperature-scaled dimensionless observable is

$$\mathcal{O}_{\text{scaled}} = 16 \times T \times \Delta_{l,s} \quad (3)$$

1.2 Chiral Susceptibility

The light-quark disconnected chiral susceptibility χ is defined as

$$\chi = \frac{V}{T} \left[\left\langle \left(\frac{1}{V} \text{Tr}(D_l^{-1}) \right)^2 \right\rangle - \left\langle \frac{1}{V} \text{Tr}(D_l^{-1}) \right\rangle^2 \right] \quad (4)$$

Using k stochastic volume noise vectors $O_{l,i}$ on each gauge configuration without vector-level RM subtraction, the noise bias is removed by the unbiased quadratic cross-product estimator:

$$\bar{O} = \frac{1}{k} \sum_{i=1}^k O_{l,i}, \quad \overline{O^2} = \frac{1}{k(k-1)} \sum_{i \neq j} O_{l,i} O_{l,j} = \frac{1}{k(k-1)} \left[\left(\sum_{i=1}^k O_{l,i} \right)^2 - \sum_{i=1}^k O_{l,i}^2 \right] \quad (5)$$

The ensemble susceptibility and its Jackknife statistical error are given by

$$a_r = \text{JK}(\bar{O})_r, \quad b_r = \text{JK}(\overline{O^2})_r, \quad \chi_r = b_r - a_r^2 \quad (6)$$

$$\bar{\chi} = \frac{1}{N} \sum_{r=1}^N \chi_r, \quad \sigma_\chi = \sqrt{\frac{N-1}{N} \sum_{r=1}^N (\chi_r - \bar{\chi})^2} \quad (7)$$

The continuous thermal scaled susceptibility with physical dimensions of [MeV²] is

$$\chi_{\text{scaled}} = (16T)^2 a^2 \chi = N_s^3 \times N_t^3 \times T^2 \times \chi \quad [\text{MeV}^2] \quad (8)$$

2 Chiral Condensate Data and Temperature Dependence

2.1 Numerical Data Tables

Table 1 lists the simulation parameters and measured chiral condensates for the standard temperature sequence on $48^3 \times 16$ lattices. Table 2 summarizes the temporal extent scaling ensembles at $\beta = 4.17$.

Table 1: Standard $48^3 \times 16$ temperature sequence: lattice parameters, bare condensates, and renormalized subtracted condensate $\Delta_{l,s}$ (Jackknife errors in parentheses).

β	T [MeV]	m_l	m_s	m_{res}	Z_m	N_{cfg}	$\langle\bar{\psi}\psi\rangle_l$	$\Delta_{l,s}$
4.13	138.23	0.000805	0.043547	7.31×10^{-4}	0.9377	2000	$1.6688(63) \times 10^{-3}$	$1.0608(67) \times 10^{-3}$
4.15	145.75	0.000930	0.040843	4.98×10^{-4}	0.9524	2024	$1.3591(64) \times 10^{-3}$	$7.7109(67) \times 10^{-4}$
4.17	153.31	0.001001	0.038400	3.40×10^{-4}	0.9662	2008	$1.0740(64) \times 10^{-3}$	$5.0728(66) \times 10^{-4}$
4.18	157.03	0.001022	0.037265	2.80×10^{-4}	0.9729	2029	$9.7061(64) \times 10^{-4}$	$4.1649(66) \times 10^{-4}$
4.20	164.55	0.001041	0.035150	1.91×10^{-4}	0.9857	2072	$7.6708(55) \times 10^{-4}$	$2.3896(56) \times 10^{-4}$
4.23	176.43	0.001033	0.032315	1.08×10^{-4}	1.0038	2012	$5.8018(38) \times 10^{-4}$	$9.2886(38) \times 10^{-5}$
4.30	202.52	0.000939	0.026930	2.81×10^{-5}	1.0422	2052	$4.1210(14) \times 10^{-4}$	$8.2043(13) \times 10^{-6}$
4.405	241.60	0.000760	0.021032	3.75×10^{-6}	1.0927	4859	$3.1110(1) \times 10^{-4}$	$3.4370(1035) \times 10^{-7}$

Table 2: Finite temporal extent scaling ensembles at $\beta = 4.17$ with $m_l = 0.0020, m_s = 0.0400$ ($T = 153.31 \times 16/N_t$ [MeV]).

Ensemble	Geometry	T [MeV]	N_{cfg}	$\langle\bar{\psi}\psi\rangle_l$	$\langle\bar{\psi}\psi\rangle_s$	$\Delta_{l,s}$
L48T18 (stream 1)	$48^3 \times 18$	136.28	98	$1.63(8) \times 10^{-3}$	$1.42(7) \times 10^{-2}$	$8.30(44) \times 10^{-4}$
L48T18 (stream 2)	$48^3 \times 18$	136.28	76	$9.02(514) \times 10^{-5}$	$2.31(231) \times 10^{-4}$	$7.95(466) \times 10^{-5}$
L40T16	$40^3 \times 16$	153.31	879	$4.14(85) \times 10^{-5}$	$1.59(56) \times 10^{-4}$	$3.33(74) \times 10^{-5}$
L32T14	$32^3 \times 14$	175.21	291	$3.47(347) \times 10^{-6}$	$6.00(600) \times 10^{-5}$	$-1.15(115) \times 10^{-8}$
L32T12	$32^3 \times 12$	204.41	108	$6.86(43) \times 10^{-4}$	$1.22(7) \times 10^{-2}$	$-2.05(189) \times 10^{-5}$

2.2 Plots vs Temperature

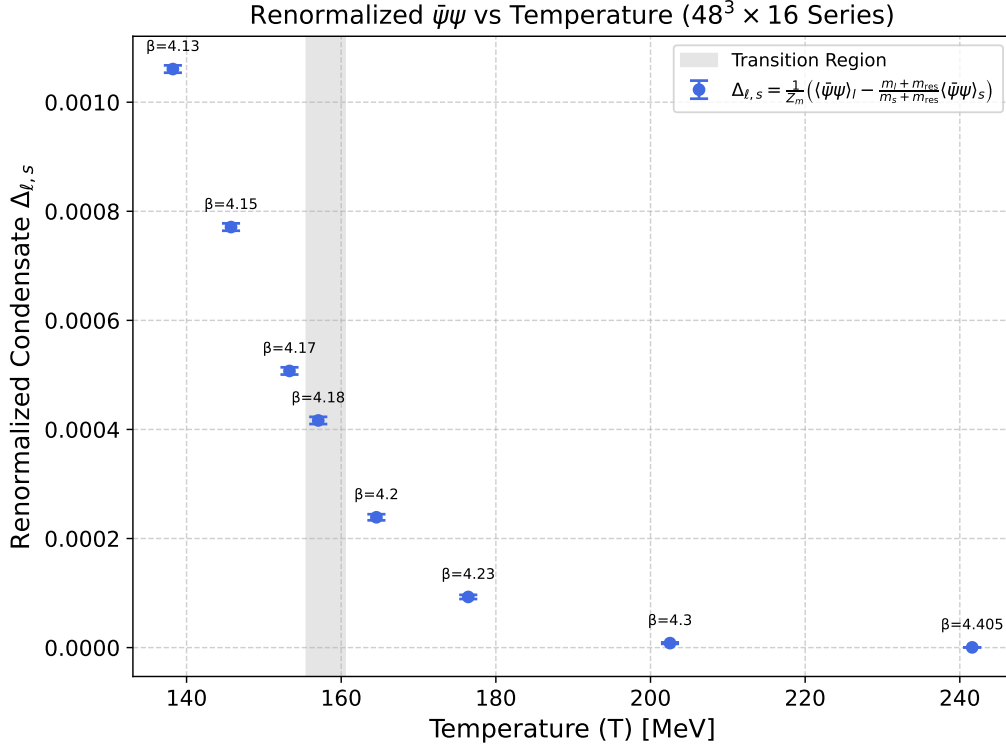
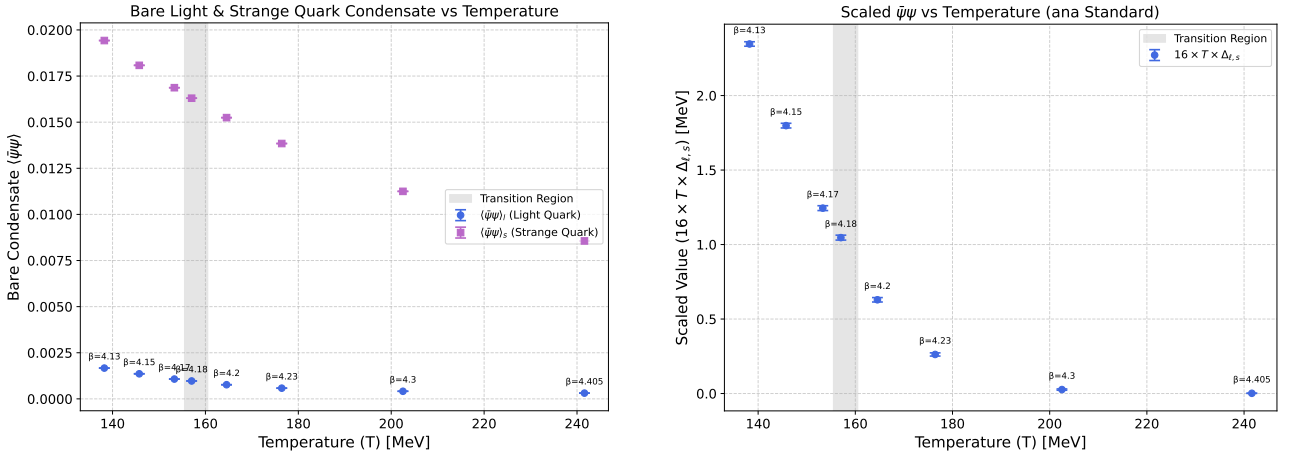


Figure 1: Renormalized subtracted chiral condensate $\Delta_{l,s}$ as a function of temperature T [MeV] for the standard $48^3 \times 16$ series. The shaded band marks the pseudo-critical transition window $T \in [155.5, 160.5]$ MeV.



(a) Bare light and strange quark condensates

(b) Temperature-scaled condensate $16T\Delta_{l,s}$

Figure 2: (a) Bare condensates $\langle \bar{\psi}\psi \rangle_l$ (blue circles) and $\langle \bar{\psi}\psi \rangle_s$ (purple squares) vs temperature T . (b) Temperature-scaled condensate $16T\Delta_{l,s}$ vs temperature T .

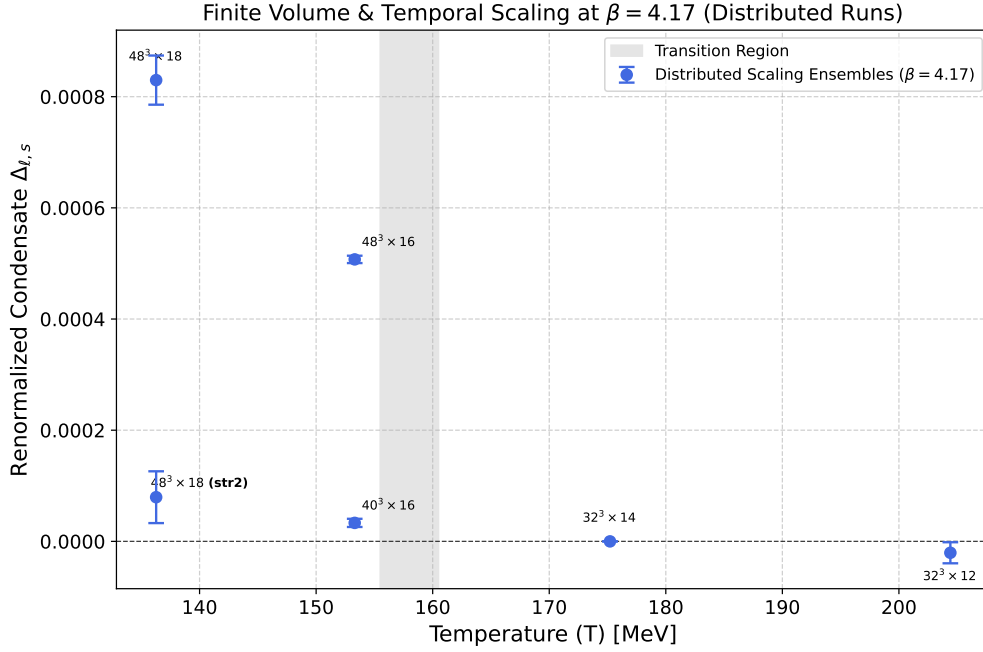


Figure 3: Subtracted chiral condensate across temporal geometries $N_t = 12, 14, 16, 18$ ($T = 153.31 \times 16/N_t$ [MeV]) at fixed $\beta = 4.17$.

3 Chiral Susceptibility Data and Temperature Dependence

3.1 Numerical Data Tables

Table 3 presents the continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ across the 8 standard temperature points ($48^3 \times 16$). Table 4 lists the chiral susceptibility values for the finite temporal scaling ensembles at $\beta = 4.17$.

Table 3: Continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ on $48^3 \times 16$ lattices (unbiased quadratic estimator without RM subtraction).

β	T [MeV]	Geometry ($N_s^3 \times N_t$)	N_{cfg}	$\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV ²]
4.13	138.23	$48^3 \times 16$	2000	$(6.5995 \pm 0.2299) \times 10^5$
4.15	145.75	$48^3 \times 16$	2024	$(7.7201 \pm 0.2646) \times 10^5$
4.17	153.31	$48^3 \times 16$	2008	$(8.5481 \pm 0.2567) \times 10^5$
4.18	157.03	$48^3 \times 16$	2029	$(9.1460 \pm 0.3228) \times 10^5$
4.20	164.55	$48^3 \times 16$	2072	$(7.6549 \pm 0.2767) \times 10^5$
4.23	176.43	$48^3 \times 16$	2012	$(4.0707 \pm 0.2304) \times 10^5$
4.30	202.52	$48^3 \times 16$	2052	$(7.1552 \pm 1.4950) \times 10^4$
4.405	241.60	$48^3 \times 16$	4859	$(8.6496 \pm 1.6936) \times 10^1$

Table 4: Chiral susceptibility χ_{scaled} for finite temporal extent ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

Ensemble	Geometry	T [MeV]	N_{cfg}	$\chi_{\text{scaled}} [(16T)^2 a^2 \chi]$ [MeV ²]
L48T18 (stream 1)	$48^3 \times 18$	136.28	98	To be populated upon remote scan
L48T18 (stream 2)	$48^3 \times 18$	136.28	76	To be populated upon remote scan
L40T16	$40^3 \times 16$	153.31	879	To be populated upon remote scan
L32T14	$32^3 \times 14$	175.21	291	To be populated upon remote scan
L32T12	$32^3 \times 12$	204.41	108	$(4.4353 \pm 1.8893) \times 10^5$

3.2 Plots vs Temperature

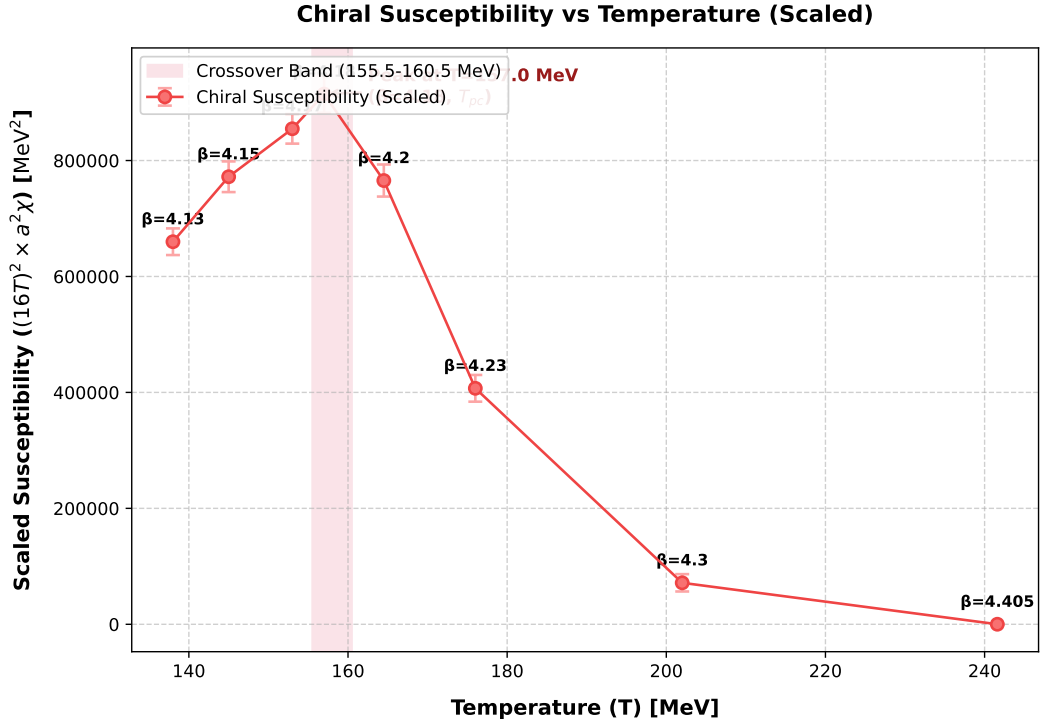


Figure 4: Continuous scaled chiral susceptibility $\chi_{\text{scaled}} = (16T)^2 a^2 \chi$ [MeV²] as a function of temperature T [MeV]. The peak marks the pseudo-critical temperature $T_{\text{pc}} = 157.03$ MeV ($\beta = 4.18$). The shaded band marks the transition window $T \in [155.5, 160.5]$ MeV.

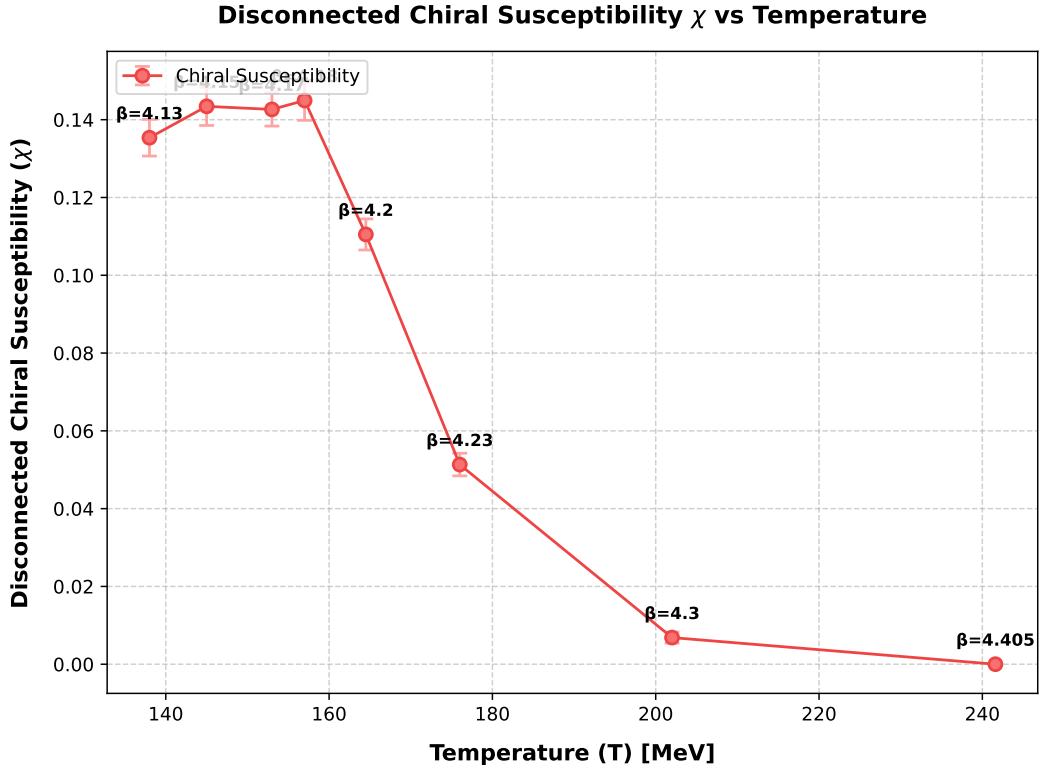


Figure 5: Four-dimensional volume-scaled chiral susceptibility $\chi_{\text{vol}} = N_s^3 N_t \chi$ as a function of temperature T [MeV] for $48^3 \times 16$ lattices.

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